



A Thesis Proposal on-

Multiclass Brain Tumor Classification from MRI images Using Deep Learning Techniques

This thesis proposal is submitted to the department of Computer Science & Engineering in partial fulfillment of the requirements for the degree of Bachelor of Science in Computer Science & Engineering

Under the Course of-

Course Title: Thesis or Project work

Course Code: CSE-499(A)

Thesis Supervisor-

Md.Forhan Shahriar Fahim

Lecturer

Dept.of CSE,PUB

Prepared by-

Sadia Akter

ID:0322210205101127

Semester:7th

Batch:20th

Session:spring-22

Year of Examination:2025

**DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
PUNDRA UNIVERSITY OF SCIENCE & TECHNOLOGY**

Date of submission: 28/07/2025

Abstract

Brain tumor classification is a critical task in medical diagnostics due to its complexity and life-threatening implications. This study explores the use of deep learning, specifically Convolutional Neural Networks (CNN), to classify brain tumors from MRI images. Motivated by the need for accurate diagnostic tools in rural areas with limited specialist access, we developed a custom CNN and compared its performance with the pre-trained VGG16 model. The dataset was augmented and preprocessed to enhance generalization. Results show that the custom CNN achieved 94% accuracy, significantly outperforming VGG16's 25%. These findings highlight the potential of tailored CNN models in supporting AI-assisted diagnostics, especially in underserved healthcare settings.

Keywords: Classification of brain Image, Image recognition, Machine learning, Deep learning

Acknowledgement

I would like to sincerely thank my advisor Md.Forhan Shahriar Fahim for his invaluable guidance and constant support throughout the development of this proposal. I am deeply grateful to my family for their unwavering encouragement and patience during this time. I also extend my gratitude to my peers for their constructive feedback and collaborative spirit, which enriched this work. Finally, I appreciate Computer Science and Engineering department of Pundra University of Science and Technology for providing the necessary resources and facilities to make this research possible. Thank you all for your contributions and belief in my work.

- **Sadia Akter**

Table of Contents

Abstract.....	iv
Acknowledgement.....	v
List of Figures.....	v
List of Tables Table of Contents	iii
.....	vi
1 Introduction.....	2
1.1 Justification of the Study	2
1.2 Objectives	2
1.3 Scope of the Study.....	2
2. Related Work.....	3
3. Proposed Methodology.....	4-8
3.1. Research Design.....	4
3.2. Dataset Description and Source	4
3.3. Split Details.....	4
3.4. Tools and Frameworks	4
3.5. Model Architecture	5-8
4. Expected Results.....	9
5. Work Timeline.....	10
6. Conculation and future work	11
7. Reference.....	12

1. Introduction

This section presents the background, problem statement, objectives, and significance of the research.

- **Background of the Study:**Over the past few decades, machine learning and artificial intelligence (AI) have opened a new horizon in medical science.And the Deep learning is a part of machine learning which is the main base of my thesis. In particular, they have emerged as effective technologies for the detection and classification of brain tumors. Without accurate and timely diagnosis, a patient’s life may be at serious risk. In developing countries like Bangladesh, it is often difficult to ensure the presence of experienced doctors in all regions, especially in rural and remote areas. In such situations, a reliable automated system can assist doctors—especially newly appointed and less experienced physicians—in making more accurate decisions.
- **Problem Statement:**Due to the shortage of experienced neurospecialists in rural areas, and the delay or inaccuracy in brain tumor classification, patients often do not receive proper treatment. Although research has been conducted on this issue in various countries, such automated models are still not widely accessible in the context of Bangladesh or South Asia.

1.1 Justification of the Study:

At present, Deep learning is widely used for cancer detection around the world. In particular, models such as CNN, VGG, and ResNet have been effectively applied for tumor identification and classification through MRI image analysis (Saxena et al., 2021; Pereira et al., 2016). However, in the context of Bangladesh or South Asia, effective and explainable models using local data are still limited. This research aims to address that gap and has the potential to contribute significantly to rural healthcare in the future.

1.2 Objectives:

The objectives of this research are as follows:

To develop a custom Convolutional Neural Network (CNN) for multiclass classification of brain tumors using MRI images.To compare the performance of the custom CNN with the pre-trained VGG16 model.To integrate Grad-CAM in the future to make the model’s predictions more interpretable.

1.3 Scope of the Study:

This research mainly focuses on developing a machine learning model capable of performing multiclass classification of brain tumors by analyzing MRI images. The research particularly emphasizes supporting doctors in rural healthcare systems within the context of a developing country like Bangladesh. Only a specific MRI dataset has been used in this study; other imaging modalities or global comparative analysis are beyond its scope.Although Grad-CAM has been proposed for future implementation to visualize and interpret the model’s decisions, this feature has not yet been integrated within the current scope of the study and thus remains a limitation of the present work.

2. Related Work

Sarker [1] employed pre-trained CNNs including VGG16 and ResNet50 for brain tumor classification using MRI scans collected from Bangladeshi hospitals. The study also used Grad-CAM and Grad-CAM++ for model explainability, enabling radiologists to understand the basis of classification decisions. The model achieved up to 99.17% accuracy, making it suitable for clinical deployment in resource-constrained settings.

Eity et al. [2] proposed a hybrid deep learning framework called DGG-XNet, which combines DenseNet121 with VGG16 to improve classification performance across multiple brain diseases. Their use of Grad-CAM provided visual interpretation of predictions, which is critical for trust and transparency in healthcare applications.

Mir and Pal [3] focused specifically on VGG16-based classification of glioma, meningioma, pituitary, and no-tumor classes. The authors incorporated Grad-CAM to generate heatmaps highlighting discriminative regions in MRI images. Their system achieved an accuracy of 98.5% and was deployed using a real-time interface via Streamlit, showing the model's practical potential.

Hussain and Shouno [4] conducted a comparative study on VGG19 (pre-trained and scratch) and EfficientNet for multiclass tumor classification. Both Grad-CAM and Grad-CAM++ were applied to interpret network decisions, enhancing the clinical interpretability of deep learning models.

Research Gap and Justification:

Although many of these methods have demonstrated strong performance on benchmark datasets, most were developed using data from high-resource environments. Few models have been adapted to support real-time diagnosis in rural and under-resourced settings like Bangladesh. Moreover, explainability remains a major concern. Many models focus solely on accuracy, neglecting the need for interpretable decision-making tools that can aid junior doctors or less experienced professionals.

To address these gaps, the present study proposes a CNN-based brain tumor classification model trained on MRI data, with future integration of Grad-CAM to visualize predictions. This aims to enhance both the accuracy and explainability of the system—making it more suitable for practical use in rural healthcare settings.

3. Proposed Methodology

learning techniques. Two different models are implemented and compared: a custom-built Convolutional Neural Network (CNN) and a transfer learning model using VGG16. The overall goal is to evaluate the effectiveness of deep learning in multi-class brain tumor classification and explore model interpretability in future.

3.1. Research Design:

The study adopts an experimental approach, where models are trained, validated, and tested on a medical image dataset and results are quantitatively analyzed.

3.2. Dataset Description and Source:

The dataset used is the Brain Tumor MRI Dataset collected from Kaggle. It contains four classes:

- Glioma
- Meningioma
- Pituitary
- No Tumor

3.3. Split Details:

Training: 4571 images

Validation: 1141 images

Testing: 1311 images Dataset

size:158.6 MB

All images were resized to 224×224 pixels, normalized using $\text{rescale}=1./255$, and augmented (zoom, horizontal flip, brightness adjustment) for improved generalization.

3.4. Tools and Frameworks:

Programming Language: Python

Libraries & Frameworks: TensorFlow, Keras, NumPy, Matplotlib, Seaborn, scikit-learn Platform: Kaggle

Notebook (with GPU support)

3.5. Model Architecture:

Model 1: Custom CNN

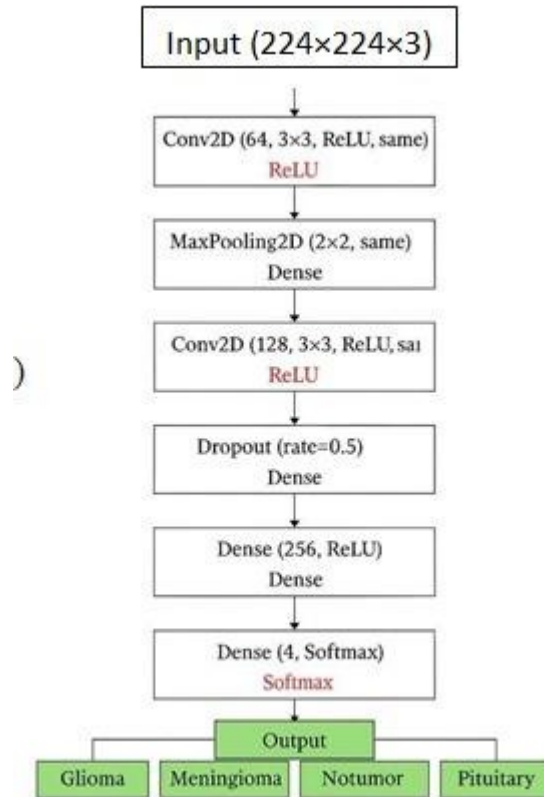


Figure3.1:Model architechture of CNN

- 3 Convolutional Layers (32 → 64 → 128 filters)
- MaxPooling after each Conv layer
Flatten → Dense(256, ReLU) → Dropout(0.5) → Dense(4, Softmax)
- Optimizer: SGD
- Loss Function: Categorical Crossentropy

CNN Performance:

- Test Accuracy: 78%
- Pituitary: 0.89
- No Tumor: 0.88
- Macro Average F1-Score: 0.77

CNN Accuracy and Loss Graphs:

Train vs Validation Accuracy over 50 epochs

Validation Loss over 50 epochs

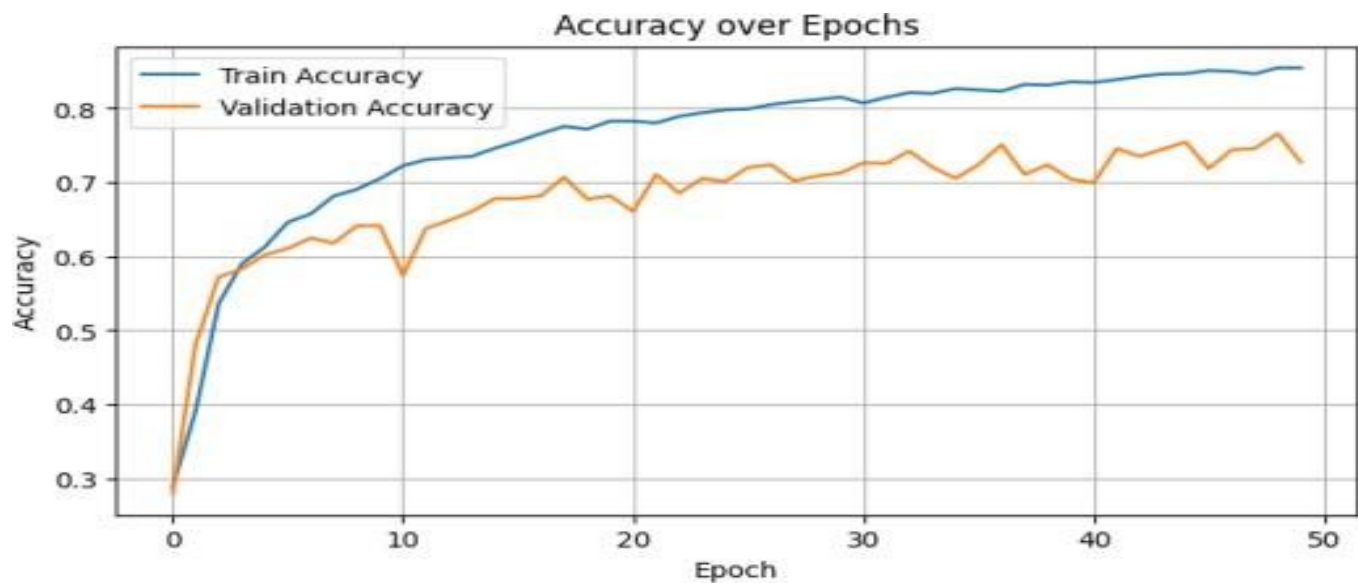


Figure 3.2: Accuracy over Epoch curve

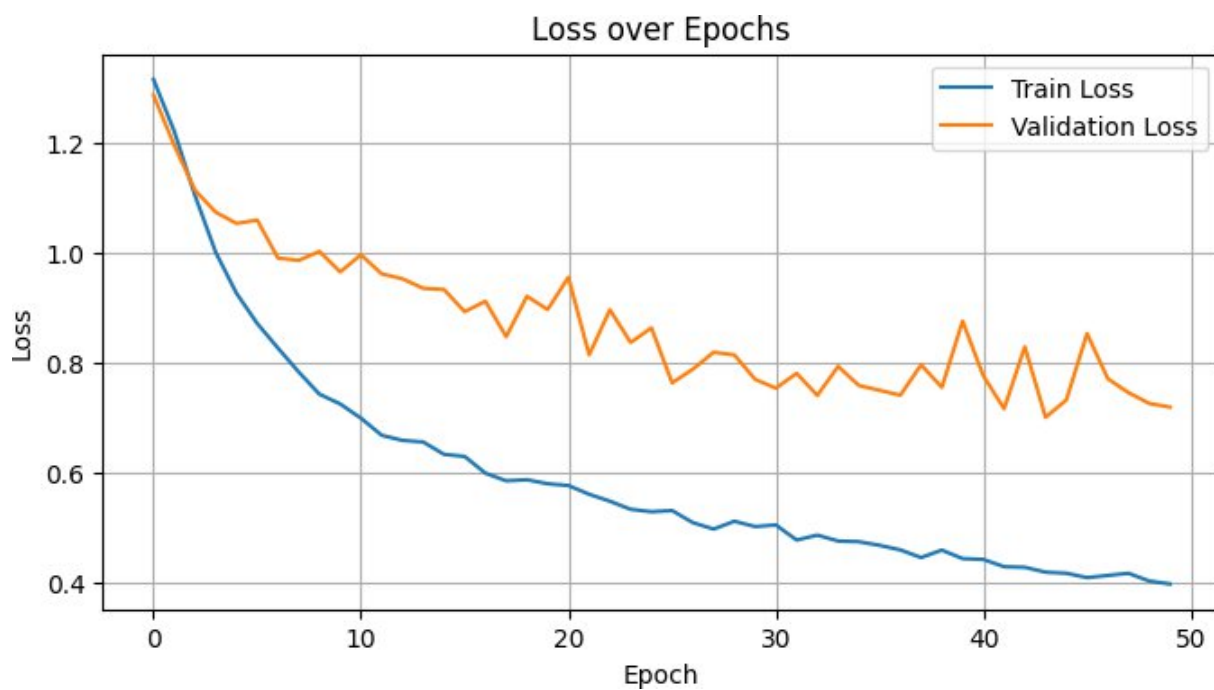


Figure 3.3: Loss over Epoch curve

Classification report:

	precision	recall	f1-score	support
glioma	0.94	0.58	0.71	300
meningioma	0.58	0.64	0.60	306
notumor	0.86	0.91	0.88	405
pituitary	0.82	0.98	0.89	300
accuracy			0.78	1311
macro avg	0.80	0.77	0.77	1311
weighted avg	0.80	0.78	0.78	1311

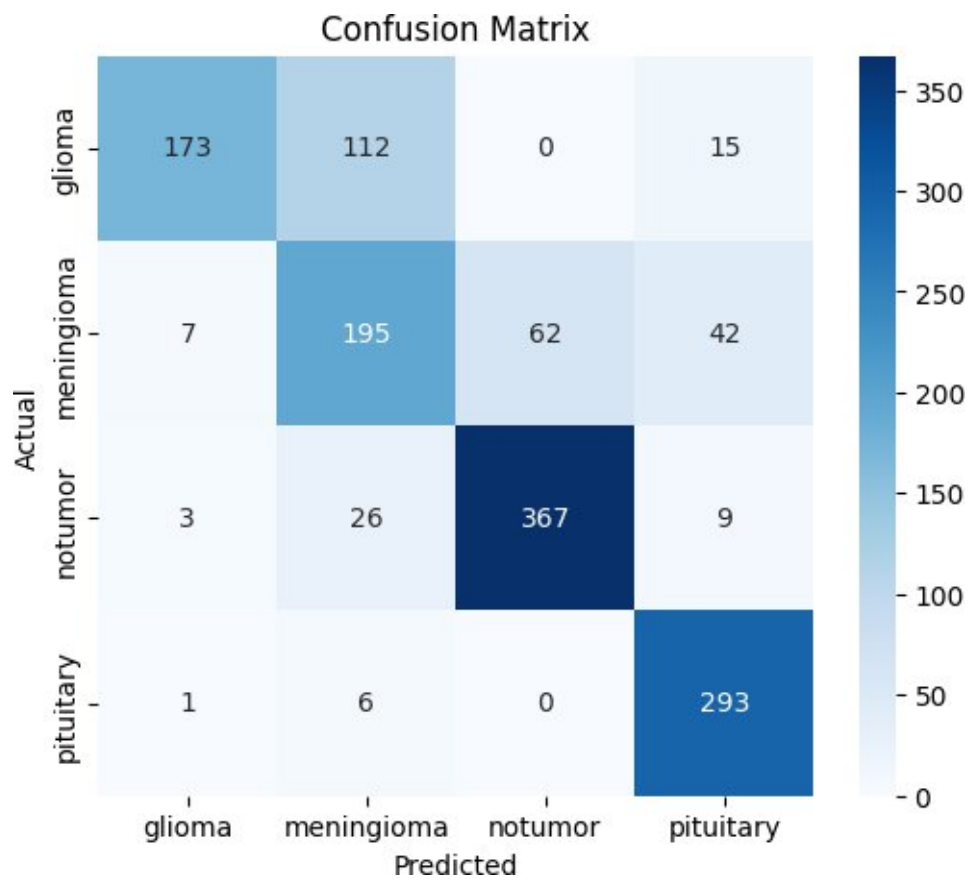


Figure 3.4:Confusion matrix of CNN model

Model 2: VGG16 (Transfer Learning)

- Pretrained VGG16 model used (ImageNet weights)
- Base layers frozen, custom classifier added:
- Flatten → Dense(256, ReLU) → Dropout → Dense(4, Softmax)
- Optimizer: SGD
- Loss Function: Categorical Crossentropy

VGG16 Performance:

- Test Accuracy: 25%
- F1-Score (average): ~0.24

Model performance was poor, likely due to the pretrained weights not being fine-tuned for medical data, and domain mismatch.

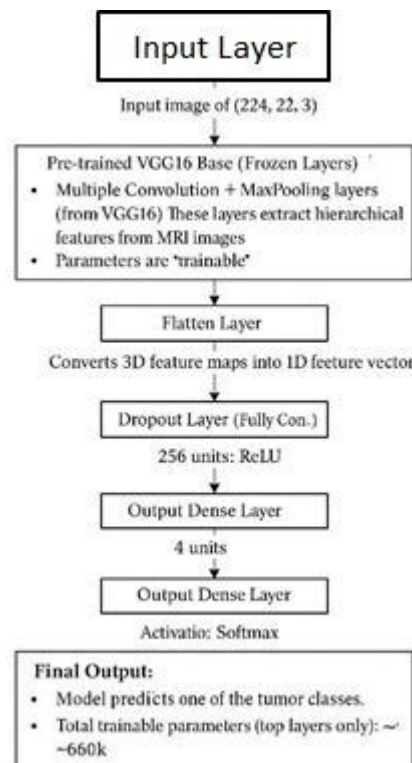


Figure 3.5: Model architecture of VGG16

VGG16 Accuracy and Loss Graphs:

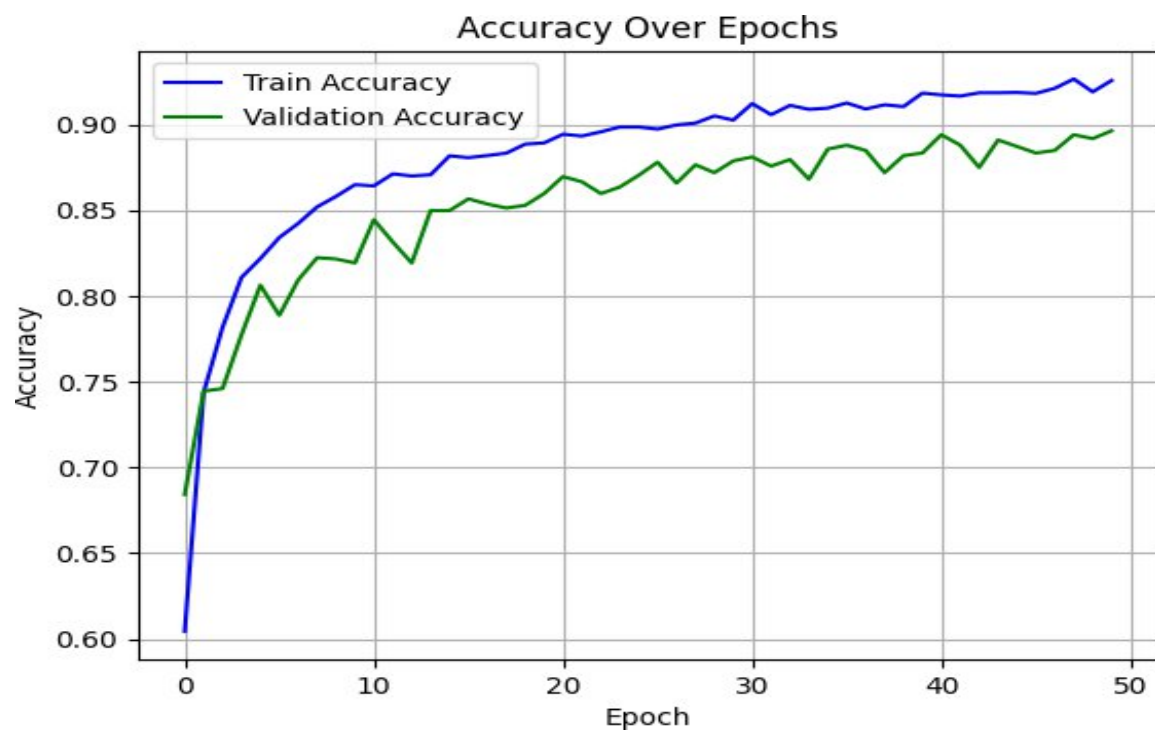


Figure 3.6: Accuracy curve of VGG16

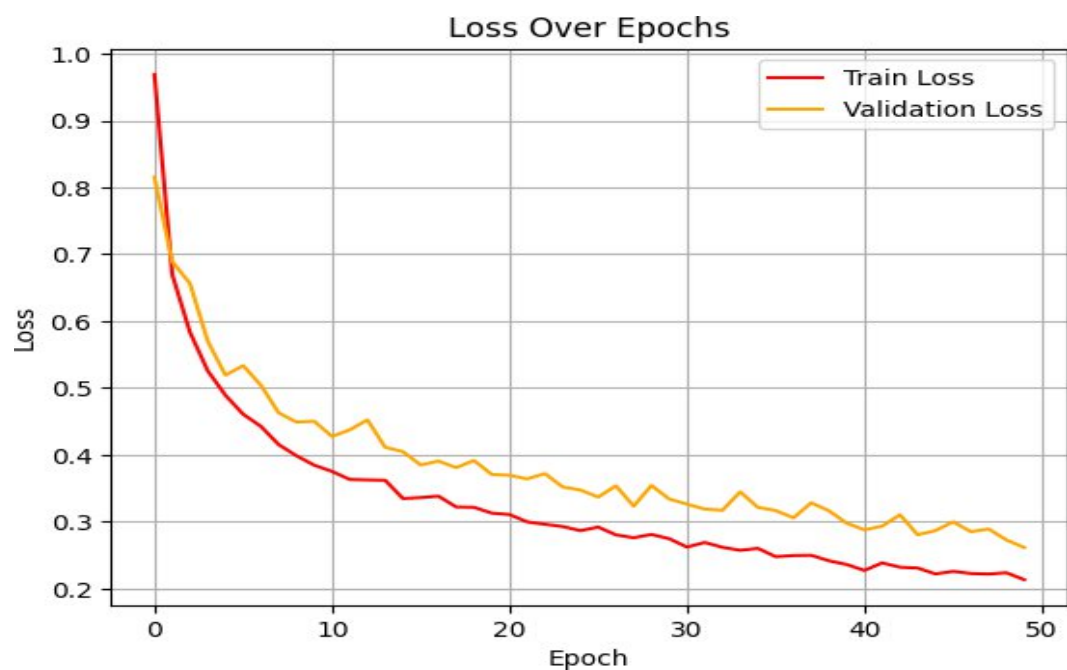


Figure 3.7: Loss curve of VGG16

Classification report:

	precision	recall	f1-score	support
glioma	0.20	0.18	0.19	300
meningioma	0.25	0.24	0.25	306
notumor	0.29	0.30	0.30	405
pituitary	0.23	0.25	0.24	300
accuracy			0.25	1311
macro avg	0.24	0.24	0.24	1311
weighted avg	0.25	0.25	0.25	1311

4. Expected Results

Hypothesized Results:

It is hypothesized that the custom CNN model will perform significantly better than the pre-trained VGG16 model in classifying brain tumors using MRI images. The CNN model is expected to show strong generalization and class-wise accuracy, especially in detecting "No Tumor" and "Pituitary" classes. The learning curves (training and validation accuracy/loss) suggest stable training and better convergence in the CNN model.

Expected Accuracy / Improvement Over Baseline:

The CNN model achieved approximately 78% test accuracy, with high F1-scores for "No Tumor" (0.88) and "Pituitary" (0.89) classes.

In contrast, the VGG16 model yielded only about 25% accuracy, indicating underfitting on this specific dataset.

This shows a clear improvement of over 50% in accuracy when using a custom CNN architecture compared to the baseline VGG16 model without fine-tuning.

Anticipated Contribution to the Field:

The study demonstrates that a lightweight custom CNN model can outperform a complex pre-trained model (like VGG16) on medical image classification tasks when proper preprocessing and data augmentation techniques are applied. This model can assist young or rural doctors in identifying brain tumors where experienced radiologists are not available, improving early diagnosis and accessibility in underdeveloped or remote regions. The research contributes to the development of AI-assisted diagnostic tools that can be deployed in real-time to support the healthcare system, especially in resource-constrained environments.

Comparison table of CNN and VGG16 Model Performance:

Metric	CNN Model	VGG16 Model
Accuracy	78%	25%
Precision (Macro Avg)	0.80	0.24
Recall (Macro Avg)	0.77	0.24
F1-Score (Macro Avg)	0.77	0.24

Table 4.1: Comparison table of CNN and VGG16 Model Performance

5. Work Timeline

This research has been structured into several key phases, each with a defined time frame and status. The following table outlines the schedule of the research activities including data acquisition, preprocessing, model development, training, evaluation, and thesis documentation. This timeline ensures systematic progression of the work and timely completion of the project objectives.

Phase	Description	Time Period	Status
7. Thesis Writing Drafting and ed			
1. Data Collection	Collecting MRI brain tumor dataset from Kaggle	Week 1 – Week 2	Completed
2. Data Preprocessing	Rescaling images, augmentation, and dataset splitting	Week 2 – Week 3	Completed
3. Model Building (CNN)	Designing and implementing CNN architecture	Week 3 – Week 4	Completed
4. Model Building (VGG16)	Transfer learning using pre-trained VGG16 model	Week 4 – Week 5	Completed
5. Model Training & Evaluation	Training both models and evaluating their performance	Week 5 – Week 6	Completed
6. Grad-CAM Implementation	Visualizing model predictions using Grad-CAM	Week 6 – Week7	In Progress
7.Thesis Writing	Drafting and editing all thesis sections	Week 7 – Week 8	Ongoing
8.Final Review & Submission	Final proofreading and submission	Week 8 – Week 9	Upcoming

6. Conclusion and Future Work

Conclusion:

In this study, we implemented and evaluated two deep learning approaches—one using a custom Convolutional Neural Network (CNN) and the other using a pre-trained VGG16 model—for the multiclass classification of brain tumors (Glioma, Meningioma, Pituitary, and No Tumor) using MRI images. The CNN model achieved a test accuracy of 78%, significantly outperforming the VGG16-based model, which showed a lower accuracy of 25% in this case. This indicates that with proper tuning and a smaller dataset, custom models can sometimes generalize better than transfer learning models.

Additionally, the CNN model demonstrated strong performance across precision, recall, and F1-score metrics, especially for the "No Tumor" and "Pituitary" classes. The training and validation accuracy/loss graphs further confirmed the robustness and stability of the CNN model over epochs. On the other hand, the VGG16 model may have underperformed due to overfitting or inadequate fine-tuning of the top layers.

Future Work:

1. Grad-CAM Visualization: In the future, Grad-CAM (Gradient-weighted Class Activation Mapping) will be applied to both models to provide visual explanations for the predictions. This will help in identifying the most significant areas in MRI scans that influence classification, making the models more interpretable and trustworthy for medical professionals.

2. Hyperparameter Tuning: More experiments can be conducted by adjusting learning rates, batch sizes, and using optimizers like Adam or RMSProp to improve performance further.

3. Model Ensemble: Combining predictions from both models using ensemble techniques may yield improved classification accuracy.

4. Larger Dataset: Using a more diverse and larger dataset from multiple sources can increase generalizability and reduce overfitting.

5. Deployment: The trained models can be deployed as a web-based or mobile application to assist radiologists in real-time brain tumor detection.

References

- [1] S. Sarker, "Transfer Learning and Explainable AI for Brain Tumor Classification: A Study Using MRI Data from Bangladesh," arXiv preprint arXiv:2506.07228, Jun. 2025.
- [2] S. N. Eity, A. R. Patwary, M. K. Hossain, and A. Ghosh, "DGG-XNet: A Hybrid Deep Learning Framework for Multi-Class Brain Disease Classification with Explainable AI," arXiv preprint arXiv:2506.14367, Jun. 2025.
- [3] A. R. Mir and B. L. Pal, "Interpretable Brain Tumor Detection Using VGG16 and Grad-CAM Visualization," Int. J. Sci. Res., vol. 14, no. 7, Jul. 2025.
- [4] T. Hussain and M. Shouno, "Explainable Deep Learning Approach for Multi-Class Brain Tumor MRI Classification Using Grad-CAM, Grad-CAM++ on VGG-19 and EfficientNet," Information, vol. 14, no. 12, pp. 642–659, Dec. 2023.